convenia®

(cefovecin sodium)

Convenia 80 mg/ml Powder for Solution for Injection for Dogs and Cats

DOSAGE FORM

Powder for solution for injection.

DESCRIPTION

The product is presented as an off-white to yellow lyophilized powder for injection, intended to be reconstituted with Water for Injection. Each ml of Convenia reconstituted lyophile contains cefovecin sodium equivalent to 80 mg cefovecin , methylparaben 1.8 mg (preservative), propylparaben 0.2 mg (preservative), sodium citrate dihydrate 5.8 mg and citric acid monohydrate 0.1 mg, sodium hydroxide or hydrochloric acid as required to adjust pH.

Cefovecin sodium is a semi-synthetic broad-spectrum antibacterial agent from the cephalosporin class of chemotherapeutic agents.
Cefovecin is the non-proprietary designation for (6*R*,7*R*)-7-[[(2*Z*)-(2-amino-4-thiazolyl) (methoxyimino) acetyl]amino]-8-oxo-3-[(2*S*)-tetrahydro-2-furanyl]-5-thia-1-azabicyclo [4.2.0]oct-2-ene-2-carboxylic acid, monosodium salt.

Reconstituted solution:
The reconstituted solution is prepared by injecting Water for Injection into the vial containing the powder for injection. The solution is visually free of undissolved matter and essentially free from foreign matter. It is light yellow to reddish-brown in colour.

PHARMACODYNAMICS

Cefovecin is a cephalosporin antibiotic. Like other β-lactam antimicrobials, Cefovecin exerts its inhibitory effect by interfering with bacterial cell wall synthesis. This interference is primarily due to its covalent binding to the penicillin-binding proteins (PBPs) (i.e., transpeptidase and carboxypeptidase), which are essential for synthesis of the bacterial cell wall.

PHARMACOKINETICS

Cefovecin is rapidly absorbed, with peak plasma concentrations of the drug attained approximately 2 hours post-dose. This drug exhibits non-linear pharmacokinetics. It is fully bioavailable following subcutaneous (SC) administration.
The pharmacokinetic parameters (mean ± standard deviation) for Cefovecin following 8mg/kg administered subcutaneous administered to dogs and cats are as follows:

Parameter	**Mean ±SD**	
	Dogs	**Cats**
Half life, $t_{½}$ (hours)	133.1 ± 15.9	166.3 ± 18.2 ^{P}
AUC (SC) 1st phase* (µg.h/ml)	10416 ± 1877	22685 ± 3452
AUC (SC) 2nd phase* (µg.h/ml)	14125 ± 1490	37130 ± 3774
T_{max} (hours)	6.208 ± 3.041	2.0 ± 2.0 ^{P}
Maximum concentration, C_{max} (µg/ml)	121.1 ± 50.7	141.0 ± 11.8 ^{P}

^{P} = a phase effect was observed, only data for the 1st phase are provided (n=6)

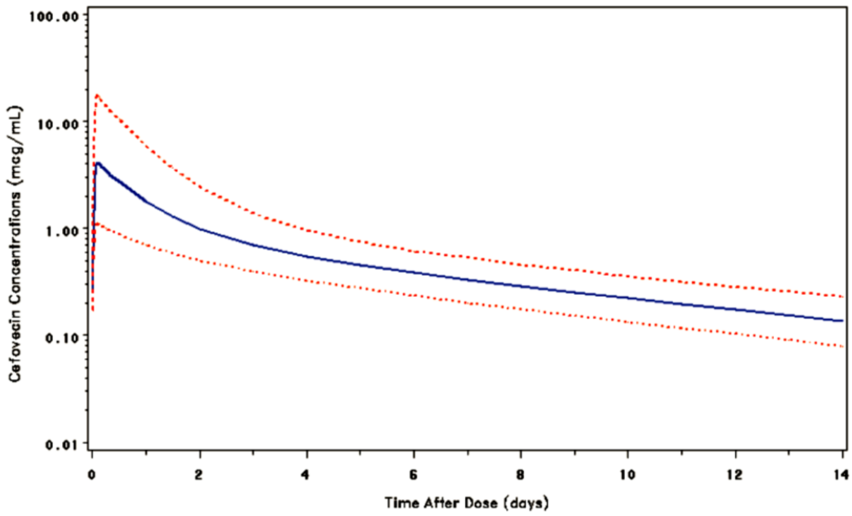
Population Pharmacokinetics in Dogs

Plasma cefovecin concentration data were pooled from seven laboratory pharmacokinetic studies for use in a PPK (population pharmacokinetics) analysis. The final dataset contained 591 concentration records from 39 dogs. The program NONMEM version 6 was used to fit various PPK models to the data and to perform parametric bootstrap simulations to evaluate the stability of the final model.

The simulations indicated that > 99% of dogs will have free plasma cefovecin concentrations ≥ 0.06 µg/mL for the 28 days after two 8-mg/kg SC doses separated by 14-days. With a 7-day dosing interval, approximately 83% of dogs are predicted to have free concentrations ≥ 0.25 µg/mL at 7-days after the 1st dose. There is accumulation of cefovecin after the 2nd dose, and a higher percentage of dogs, approximately 98%, are predicted to have free plasma cefovecin concentrations ≥ 0.25 µg/mL at 7-days after the 2nd dose. The results of these simulations are consistent with the observed efficacy of cefovecin in the treatment of bacterial infections caused by cefovecin-susceptible pathogens.

Figure 10 shows the median, 5th and 95th percentiles of the predicted free plasma cefovecin concentrations after the 1st-dose with an expanded 14-day time-scale.

Figure 10. Simulated free plasma cefovecin concentrations versus time for the 14-days after a single-dose of cefovecin 8 mg/kg SC. Key: solid curve: median predicted free concentrations, dashed curves: 5th and 95th percentiles of the distribution of predicted free concentrations.

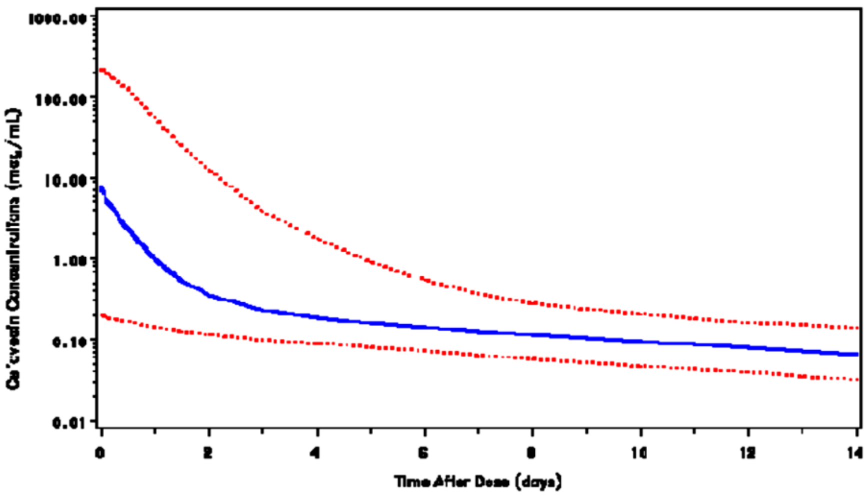


Population Pharmacokinetics in Cats

Plasma cefovecin concentration data were pooled from four laboratory pharmacokinetic studies for use in a population pharmacokinetic analysis. The final dataset contained 338 concentration records from 22 cats. The program NONMEM version 6 was used to fit various PPK models to the data and to perform parametric bootstrap simulations to evaluate the stability of the final model.

The simulations indicated that free plasma cefovecin concentrations will be ≥ 0.06 µg/mL in approximately 96% and of cats at 7 days after a single 8 mg/kg SC dose of cefovecin.

Figure 9. Simulated free plasma cefovecin concentrations versus time from a single 8 mg/kg SC dose of cefovecin. Key: solid curve: median predicted free concentrations, dashed curves: 5th and 95th percentiles of the distribution of predicted free concentrations.



INDICATIONS

Dogs

CONVENIA is indicated for the treatment of skin infections (secondary superficial pyoderma, abscesses, and wounds) in dogs caused by susceptible strains of *Staphylococcus intermedius* and *Streptococcus canis* (Group G).

Cats

CONVENIA is indicated for the treatment of skin infections (wounds and abscesses) in cats caused by susceptible strains of *Pasteurella multocida*.

RECOMMENDED DOSAGE

Dogs

CONVENIA should be administered as a single subcutaneous injection of 8 mg/kg (3.6 mg/lb) body weight. A second subcutaneous injection of 8 mg/kg (3.6 mg/lb) may be administered if response to therapy is not complete. The decision for a second injection for any individual dog should take into consideration such factors as progress toward clinical resolution, the susceptibility of the causative organisms, and the integrity of the dog's host-defense mechanisms. Therapeutic drug concentrations after the first injection are maintained for 7 days for *S. intermedius* infections and for 14 days for *S. canis* (Group G) infections. Maximum treatment should not exceed 2 injections.

Cats

CONVENIA should be administered as a single, one-time subcutaneous injection at a dose of 8 mg/kg (3.6 mg/lb) body weight. After an injection of CONVENIA, therapeutic concentrations are maintained for approximately 7 days for *Pasteurella multocida* infections.

Dose Table for CONVENIA at 8 mg/kg Body Weight

Animal Weight (Dogs and Cats)	Volume of CONVENIA
2.5 kg	0.25 ml
5 kg	0.5 ml
10 kg	1.0 ml
20 kg	2.0 ml
40 kg	4.0 ml
60 kg	6.0 ml

ROUTE OF ADMINISTRATION

Subcutaneous

CONTRAINDICATION

Cefovecin is contraindicated in dogs and cats with known allergy to cefovecin or to β-lactam (penicillins and cephalosporins) group antimicrobials.

If an allergic reaction or anaphylaxis occurs, cefovecin should not be administered again and appropriate therapy should be instituted. Anaphylaxis may require treatment with epinephrine and other emergency measures, including oxygen, intravenous fluids, intravenous antihistamine, corticosteroids, and airway management, as clinically indicated. Adverse reactions may require prolonged treatment due to the prolonged systemic drug clearance (65 days).

WARNING AND PRECAUTIONS

Not for use in humans. Keep this and all drugs out of reach of children. Jauhi dari kanak-kanak.

Consult a physician in case of accidental human exposure. For subcutaneous use in dogs and cats only. Antimicrobial drugs, including penicillins and cephalosporins, can cause allergic reactions in sensitized individuals. To minimize the possibility of allergic reactions, those handling such antimicrobials, including cefovecin, are advised to avoid direct contact of the product with the skin and mucous membranes.

INTERACTION WITH OTHER MEDICINAL PRODUCT

Dogs

When tested in an *in vitro* equilibrium dialysis system using canine plasma at therapeutically relevant concentrations, carprofen, furosemide, doxycycline and ketoconazole appear to have little impact on the plasma protein binding of cefovecin. As a result, it is unlikely that co-administration of any of these drugs with cefovecin will have a clinically relevant change on the free concentration of cefovecin *in vivo* in dogs.

Cefovecin, on the other hand, has been shown in this experimental *in vitro* system to result in an increase in the free concentration of carprofen, furosemide, doxycycline and ketoconazole. The clinical relevance of this observation is unknown.

Cats

When tested in an *in vitro* equilibrium dialysis system using feline plasma at therapeutically relevant concentrations, carprofen, furosemide, doxycycline and ketoconazole appear to have little impact on the plasma protein binding of cefovecin. As a result, it is unlikely that co-administration of any of these drugs with cefovecin will have a clinically relevant change on the free concentration of cefovecin *in vivo* in cats.

Cefovecin, on the other hand, has been shown in this experimental *in vitro* system to result in an increase in the free concentration of furosemide and ketoconazole. The plasma protein binding of carprofen appears to be unaffected by the presence of cefovecin and the free concentration of doxycycline is similarly unaffected. The clinical relevance for the cat of these observations is unknown.

PREGNANCY AND LACTATION

The safety of Convenia in dogs and cats has not been established during pregnancy and lactation.

SIDE EFFECTS

The most frequently reported adverse events are consistent with the SPC and the safety literature for the cephalosporin drug class.
In the EU, the safety profile included very rare reports of neurologic disorders, various hypersensitivity reactions, gastro-intestinal disorders and injection site reactions.

SYMPTOMS AND TREATMENT OF OVERDOSE

Repeated dosing (eight administrations) in 14-day intervals at five times the recommended dose was tolerated well in young dogs. Slight and transient injection site swellings were observed after the first and second administration. A single administration of 22.5 times the recommended dose caused transient oedema and discomfort at the injection site.

Repeated dosing (eight administrations) in 14-day intervals at five times the recommended dose was tolerated well in young cats. A single administration of 22.5 times the recommended dose caused transient oedema and discomfort at the injection site.

INSTRUCTIONS FOR USE

Preparation of Solution For Injection:

To deliver the appropriate dose**,** aseptically reconstitute CONVENIA with 10 mL Sterile Water for Injection. Shake and allow vial to sit until all material is visually dissolved. The resulting solution contains cefovecin sodium equivalent to 80 mg/mL cefovecin. CONVENIA is light sensitive. The vial should be stored in the original carton and refrigerated when not in use. Use the entire contents of the vial within 28 days of reconstitution.

STORAGE CONDITION

Store the powder and the reconstituted product in the original carton, refrigerated at 2° to 8°C. Use the entire contents of the vial within 28 days of reconstitution.

PROTECT FROM LIGHT. After each use it is important to return the unused portion back to the refrigerator in the original carton. As with other cephalosporins, the colour of the solution may vary from clear to amber at reconstitution and may darken over time. If stored as recommended, solution colour does not adversely affect potency.

PROPOSED SHELF LIFE (AS PACKAGED FOR SALE)
36 months

PROPOSED SHELF LIFE (AFTER RECONSTITUTION OR DILUTION)
Use the entire contents of the vial within 28 days of reconstitution.

PACKING

CONVENIA is supplied in 10ml glass vial.

MANUFACTURER

Zoetis Inc.
2605 East Kilgore Road,
Kalamazoo, Michigan 49007,
UNITED STATES

PRODUCT REGISTRATION HOLDER

Zoetis Malaysia Sdn Bhd (8898-V)
Lot 3.1, Level 3, Tower 7, Avenue 3, Bangsar South,
No 8 Jalan Kerinchi, 59200 Kuala Lumpur,
MALAYSIA

DATE OF REVISION

10 July 2018